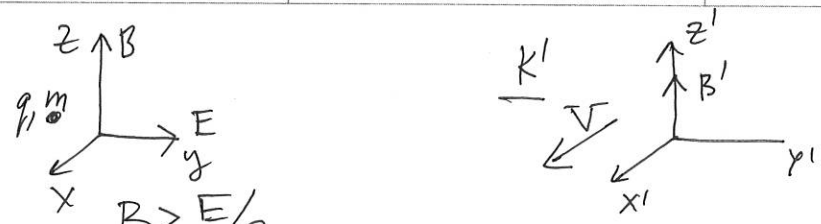
P1

K



$B > E/c$.

Frame K' moves with $\vec{V} = (V, 0, 0)$ with respect to frame K. Then in K' $E'_y = \gamma(E - vB) = 0$

$$\Rightarrow V = \frac{E}{B} < c$$

$E'_z = 0$ because $E_z = 0 = B_y$.

$$B'_z = \gamma\left(B - vE/c^2\right) = \gamma\left(B - \frac{E^2}{Bc^2}\right) = \frac{\gamma}{B}\left(B^2 - \frac{E^2}{c^2}\right).$$

But $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}} = \dfrac{1}{\sqrt{1 - \frac{E^2}{c^2B^2}}} = \dfrac{B}{\sqrt{B^2 - \frac{E^2}{c^2}}}$

$$\Rightarrow B'_z = \frac{1}{\sqrt{B^2 - E^2/c^2}}\left(B^2 - E^2/c^2\right) = \sqrt{B^2 - \frac{E^2}{c^2}}.$$

and $V = \dfrac{E}{B}$.

$\vec{E} = (0, E, 0)$, $\vec{B} = (0, 0, B)$ so that $\theta = \pi/2$ $(\cos\theta = 0, \sin\theta = 1)$

$$\frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & \Lambda & 0 \\ 0 & 0 & \omega & 0 \\ \Lambda & -\omega & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \\ u_z \end{pmatrix} \qquad \Lambda = \frac{qE}{mc} \qquad \omega = \frac{qB}{m}$$

Therefore $\frac{du_z}{d\tau} = 0$, $u_z(\tau) = u_z(\tau=0) = \frac{p_{0z}}{m}$

$$\Rightarrow z(\tau) = \frac{p_{0z}\,\tau}{m}.$$

Need to analyze 3×3 system

$$\frac{d}{d\tau}\begin{pmatrix} u_0 \\ u_x \\ u_y \end{pmatrix} = \begin{pmatrix} 0 & 0 & \Lambda \\ 0 & 0 & \omega \\ \Lambda & -\omega & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \end{pmatrix}$$

let $u \sim e^{\lambda\tau}$ $\Rightarrow$ $\lambda \vec{u} = M\vec{u} \Rightarrow (M - \lambda\hat{1})\vec{u} = 0$

$\Rightarrow$ eigenvalues are given by

$$\det\begin{pmatrix} -\lambda & 0 & \Lambda \\ 0 & -\lambda & \omega \\ \Lambda & -\omega & -\lambda \end{pmatrix} = 0 \; : \; -\lambda(\lambda^2 + \omega^2) + \Lambda(\lambda\Lambda) = 0$$

$$\Rightarrow \lambda = 0 \;, \quad \Lambda^2 - \omega^2 - \lambda^2 = 0$$

$$\lambda = \pm\sqrt{\Lambda^2 - \omega^2}$$

We have $B > \frac{E}{c} \Rightarrow \omega > \Lambda \Rightarrow \lambda = \pm i\sqrt{\omega^2 - \Lambda^2} = \pm i\Omega$

$$\Omega = \sqrt{\omega^2 - \Lambda^2} = \frac{q}{m}\sqrt{B^2 - \frac{E^2}{c^2}}$$

Eigenvectors:

$$\lambda = 0 \Rightarrow \begin{pmatrix} 0 & 0 & \Lambda \\ 0 & 0 & \omega \\ \Lambda & -\omega & 0 \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \end{pmatrix} = 0 \Rightarrow \begin{cases} \Lambda u_y = 0 \\ \omega u_y = 0 \\ \Lambda u_0 - \omega u_x = 0 \end{cases}$$

$$u_x = \frac{\Lambda}{\omega} u_0$$

$$\Rightarrow \vec{\Phi}_1 = \begin{pmatrix} 1 \\ \Lambda/\omega \\ 0 \end{pmatrix}$$

$$\lambda = i\Omega \Rightarrow \begin{pmatrix} -i\Omega & 0 & \Lambda \\ 0 & -i\Omega & \omega \\ \Lambda & -\omega & -i\Omega \end{pmatrix}\begin{pmatrix} u_0 \\ u_x \\ u_y \end{pmatrix} = 0$$

$$\begin{cases} -i\Omega u_0 + \Lambda u_y = 0 \\ -i\Omega u_x + \omega u_y = 0 \\ \Lambda u_0 - \omega u_x - i\Omega u_y = 0 \end{cases} \Rightarrow \begin{cases} u_y = \frac{i\Omega}{\Lambda} u_0 \\ u_x = \frac{\omega}{i\Omega} u_y = \frac{\omega}{\Lambda} u_0 \\ \Lambda u_0 - \frac{\omega^2}{\Lambda} u_0 - i\Omega \frac{i\Omega}{\Lambda} u_0 = 0 \end{cases}$$

$$(\Lambda^2 - \omega^2 + \Omega^2) u_0 = 0$$

okay since $\Omega^2 = \omega^2 - \Lambda^2$.

$$\vec{S}_2 = \begin{pmatrix} 1 \\ \omega/\Lambda \\ i\Omega/\Lambda \end{pmatrix}$$

$$\lambda = -i\Omega \Rightarrow \quad u_y = \frac{-i\Omega}{\Lambda} u_0$$

$$u_x = -\frac{\omega}{i\Omega} u_y = \frac{\omega}{\Lambda} u_0$$

$$\vec{S}_3 = \begin{pmatrix} 1 \\ \omega/\Lambda \\ -i\Omega/\Lambda \end{pmatrix}$$

$\Rightarrow$ general solution $\begin{pmatrix} u_0(\tau) \\ u_x(\tau) \\ u_y(\tau) \end{pmatrix} = a_p \begin{pmatrix} 1 \\ \frac{\Lambda}{\omega} \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ \frac{\omega}{\Lambda} \\ \frac{i\Omega}{\Lambda} \end{pmatrix} e^{i\Omega\tau} + a_3 \begin{pmatrix} 1 \\ \frac{\omega}{\Lambda} \\ -\frac{i\Omega}{\Lambda} \end{pmatrix} e^{-i\Omega\tau}$

Initial conditions

$$\begin{pmatrix} u_0 \\ u_x \\ u_y \end{pmatrix}_{\tau=0} = \begin{pmatrix} \varepsilon_0/mc \\ p_{0x}/m \\ p_{0y}/m \end{pmatrix} = \begin{pmatrix} a_1 + a_2 + a_3 \\ a_1 \frac{\Lambda}{\omega} + (a_2 + a_3) \frac{\omega}{\Lambda} \\ (a_2 - a_3) \frac{i\Omega}{\Lambda} \end{pmatrix}$$

Write as matrix equ:

$$K = \begin{pmatrix} 1 & 1 & 1 \\ \frac{\Lambda}{\omega} & \frac{\omega}{\Lambda} & \frac{\omega}{\Lambda} \\ 0 & \frac{i\Omega}{\Lambda} & -\frac{i\Omega}{\Lambda} \end{pmatrix} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}}_{=\vec{a}} = \frac{1}{m} \underbrace{\begin{pmatrix} \varepsilon_0/c \\ p_{0x} \\ p_{0y} \end{pmatrix}}_{=\vec{a}_0}$$

$$K\vec{a} = \vec{a}_0 \Rightarrow \vec{a} = K^{-1}\vec{a}_0 = \begin{pmatrix} \frac{\omega^2}{\Omega^2} & -\frac{\Lambda\omega}{\Omega^2} & 0 \\ -\frac{\Lambda^2}{2\Omega^2} & +\frac{\Lambda\omega}{2\Omega^2} & \frac{-i\Lambda}{2\Omega} \\ -\frac{\Lambda^2}{2\Omega^2} & \frac{\Lambda\omega}{2\Omega^2} & \frac{i\Lambda}{2\Omega} \end{pmatrix} \begin{pmatrix} \mathcal{E}_0/c \\ p_{0x} \\ p_{0y} \end{pmatrix} \frac{1}{m}$$

$$a_1 = \frac{\omega^2}{\Omega^2}\frac{\mathcal{E}_0}{mc} - \frac{\Lambda\omega}{\Omega^2}\frac{p_{0x}}{m}$$

$$a_2 = -\frac{\Lambda^2}{2\Omega^2}\frac{\mathcal{E}_0}{mc} + \frac{\Lambda\omega}{2\Omega^2}\frac{p_{0x}}{m} - \frac{i\Lambda}{2\Omega}\frac{p_{0y}}{m}$$

$$a_3 = -\frac{\Lambda^2}{2\Omega^2}\frac{\mathcal{E}_0}{mc} + \frac{\Lambda\omega}{2\Omega^2}\frac{p_{0x}}{m} + \frac{i\Lambda}{2\Omega}\frac{p_{0y}}{m}$$

This gives $u_0(\tau) = \frac{\omega^2}{\Omega^2}\left(\frac{\mathcal{E}_0}{mc} - \frac{\Lambda}{\omega}\frac{p_{0x}c}{mc}\right) +$

$$+\left(-\frac{\Lambda^2}{2\Omega^2}\frac{\mathcal{E}_0}{mc} + \frac{\Lambda\omega}{2\Omega^2}\frac{p_{0x}}{m}\right)\underbrace{\left(e^{i\Omega\tau} + e^{-i\Omega\tau}\right)}_{2\cos\Omega\tau}$$

$$-\frac{i\Lambda}{2\Omega}\frac{p_{0y}}{m}\underbrace{\left(e^{i\Omega\tau} - e^{-i\Omega\tau}\right)}_{2i\sin\Omega\tau} =$$

$$= \frac{\omega^2}{\Omega^2}\frac{\mathcal{E}_0 - \frac{\Lambda}{\omega}p_{0x}c}{mc} - \frac{\Lambda^2}{\Omega^2}\frac{\mathcal{E}_0 - \frac{\omega}{\Lambda}p_{0x}c}{mc}\cos\Omega\tau$$

$$+ \frac{\Lambda}{\Omega}\frac{p_{0y}}{m}\sin\Omega\tau \qquad (1)$$

Now, $u_0(\tau) = c\frac{dt}{d\tau} \Rightarrow ct(\tau) = \int_0^\tau d\tau\, u_0(\tau)$

$$\int d\tau \cos\Omega\tau = \frac{\sin\Omega\tau}{\Omega}, \quad \int_0^\tau d\tau \sin\Omega\tau = -\frac{\cos\Omega\tau}{\Omega}\Big|_0^\tau = \frac{1-\cos\Omega\tau}{\Omega}$$

So that $ct(\tau) = \frac{\omega^2}{\Omega^2} \frac{\mathcal{E}_0 - \frac{\Lambda}{\omega} p_{0x} c}{mc} \tau - \frac{\Lambda^2 (\mathcal{E}_0 - \frac{\omega}{\Lambda} p_{0x} c) \sin \Omega\tau}{\Omega^3 mc}$

$$+ \frac{\Lambda p_{0y}}{\Omega^2 m} (1 - \cos \Omega\tau). \quad (2)$$

Next, $u_x(\tau) = \frac{\Lambda}{\omega} a_1 + \frac{\omega}{\Lambda}\left(a_2 e^{i\Omega\tau} + a_3 e^{-i\Omega\tau}\right) =$

$$= \frac{\omega\Lambda}{\Omega^2}\left(\frac{\mathcal{E}_0}{mc} - \frac{\Lambda}{\omega}\frac{p_{0x} c}{mc}\right) + \frac{\omega}{\Lambda}\left[-\frac{\Lambda^2}{\Omega^2}\frac{\mathcal{E}_0}{mc} + \frac{\Lambda\omega}{\Omega^2}\frac{p_{0x}}{m}\right]\cos\Omega\tau$$

$$+ \frac{\omega}{\Lambda}\frac{(-i\Lambda)}{\Omega}\frac{p_{0y}}{m} i \sin\Omega\tau =$$

$$= \frac{\omega\Lambda}{\Omega^2 mc}\left(\mathcal{E}_0 - \frac{\Lambda}{\omega} p_{0x} c\right) - \frac{\omega\Lambda}{\Omega^2 mc}\left(\mathcal{E}_0 - \frac{\omega}{\Lambda} p_{0x} c\right)\cos\Omega\tau$$

$$+ \frac{\omega p_{0y}}{\Omega m}\sin\Omega\tau \quad (3).$$

Integrating over τ gives $X(\tau) = \int_0^\tau d\tau\, u_x(\tau)$

$$X(\tau) = \frac{\Lambda\omega}{\Omega^2 mc}\left(\mathcal{E}_0 - \frac{\Lambda}{\omega} p_{0x} c\right)\tau - \frac{\omega^2}{\Omega^3 mc}\left(\frac{\Lambda}{\omega}\mathcal{E}_0 - c p_{0x}\right)\sin\Omega\tau$$

$$+ \frac{\omega p_{0y}}{\Omega^2 m}(1 - \cos\Omega\tau) \quad (4).$$

Finally, $u_y(\tau) = \frac{i\Omega}{\Lambda}\left(a_2 e^{i\Omega\tau} - a_3 e^{-i\Omega\tau}\right)$

$$u_y(\tau) = \frac{i\Omega}{\Lambda}\left(-\frac{\Lambda^2}{\Omega^2}\frac{\mathcal{E}_0}{mc} + \frac{\Lambda\omega}{\Omega^2}\frac{p_{0x}}{m}\right) i \sin\Omega\tau$$

$$+ \frac{i\Omega}{\Lambda}\left(\frac{-i\Lambda}{\Omega}\right)\frac{p_{0y}}{m}\cos\Omega\tau =$$

$$= \left(\frac{\Lambda\mathcal{E}_0}{\Omega mc} - \frac{\omega p_{0x}}{\Omega m}\right)\sin\Omega\tau + \frac{p_{0y}}{m}\cos\Omega\tau =$$

$$= \frac{\omega}{\Omega mc}\left(\frac{\Lambda}{\omega}\mathcal{E}_0 - p_{0x} c\right)\sin\Omega\tau + \frac{p_{0y}}{m}\cos\Omega\tau \quad (5)$$

And $y(\tau) = \frac{\omega}{\Omega^2 mc}\left(\frac{\Lambda}{\omega}\varepsilon_0 - p_{0x}c\right)\left(1-\cos\Omega\tau\right) + \frac{p_{0y}}{m\Omega}\sin\Omega\tau$ (6)

Thus, equations (1), (3), (5) constitutes solution for $\vec{u}(\tau) =$
$= (u_0(\tau), u_x(\tau), u_z(\tau))$
while (2), (4) and (6) are $\vec{r}(\tau) = (ct(\tau), x(\tau), y(\tau)) = r^\alpha(\tau)$.

Several features of our solution are worth noting:

1) The factor $\frac{\omega}{\Omega} = \frac{\frac{qB}{m}}{\frac{q}{m}\sqrt{B^2 - \frac{E^2}{c^2}}} = \frac{B}{\sqrt{B^2 - \frac{E^2}{c^2}}} = \frac{B}{B'}$

where (according to P1) $B' = \sqrt{B^2 - E^2/c^2}$ is magnetic field in frame K' (in which $E' = 0$).

2) The factor $\frac{\Lambda}{\omega} = \frac{E}{cB} = \frac{V}{c}$ where V is velocity of K' relative to K (along x).

3) $u_x(\tau)$ has τ-independent term, which represents drift velocity of the particle in perpendicular $\vec{E}$ and $\vec{B}$ fields.

Now $\vec{E} \perp \vec{B}$ and $B = \frac{E}{c}$.

Consider $B = \frac{E}{c}\sqrt{1+\delta^2}$ so that $\Omega = \frac{q}{m}\sqrt{B^2 - \frac{E^2}{c^2}} = \frac{q\delta}{m}$

and take limit of $\delta \to 0$: remember also that

$\sin x = x - \frac{1}{6}x^3$, $1 - \cos x = \frac{1}{2}x^2$. and $\Lambda = \frac{qE}{mc} = \frac{qB}{m} = \omega$

$$ct(\tau) \to \left(\frac{m\omega}{q\delta}\right)^2 \frac{\mathcal{E}_0 - V p_{0x}}{mc}\tau - \frac{\Lambda\omega}{\left(\frac{q\delta}{m}\right)^3 mc}\left(\frac{\Lambda}{\omega}\mathcal{E}_0 - p_{0x}c\right)\cdot$$

$$\cdot\left[\Omega\tau - \frac{1}{6}\Omega^3\tau^3\right] + \frac{\Lambda p_{0y}}{m\Omega^2}\left[\frac{1}{2}\Omega^2\tau^2\right] =$$

$$= \frac{\omega^2}{\Omega^2}\frac{\mathcal{E}_0 - V p_{0x}}{mc}\tau - \frac{\Omega\tau - \frac{1}{6}\Omega^3\tau^3}{\Omega^3 mc}\left(\Lambda^2\mathcal{E}_0 - \Lambda\omega p_{0x}c\right)$$

$$+ \frac{\Lambda p_{0y}}{2m}\tau^2 = \frac{\omega^2\mathcal{E}_0 - \omega\Lambda p_{0x}c}{\Omega^2 mc}\tau$$

$$- \frac{\tau}{\Omega^2 mc}\left(\Lambda^2\mathcal{E}_0 - \Lambda\omega p_{0x}c\right) + \frac{\tau^3}{6mc}\Lambda^2\left(\mathcal{E}_0 - \frac{\omega}{\Lambda}p_{0x}c\right)$$

$$+ \frac{\Lambda p_{0y}}{2m}\tau^2 = \frac{\overbrace{(\omega^2 - \Lambda^2)}^{\Omega^2!}\mathcal{E}_0\tau}{\Omega^2 mc} + \frac{p_{0x}c\tau}{\Omega^2 mc}\underbrace{(-\omega\Lambda + \Lambda\omega)}_{\emptyset}$$

$$+ \frac{\tau^3\Lambda^2}{6mc}\left(\mathcal{E}_0 - V p_{0x}\right) + \frac{\Lambda p_{0y}}{2m}\tau^2$$

$$\Rightarrow ct = \frac{\mathcal{E}_0}{mc}\tau + \frac{qEp_{0y}}{2m^2c}\tau^2 + \frac{(qE)^2}{6(mc)^3}\left(\mathcal{E}_0 - V p_{0x}\right)\tau^3. \quad (17)$$

(Notice some remarkable cancellations that took place.)

$$x(\tau) = \frac{\Lambda\omega}{\Omega^2}\frac{\mathcal{E}_0}{mc}\left(\tau - \frac{\sin\Omega\tau}{\Omega}\right) - \frac{\Lambda^2 p_{0x}\tau}{\Omega^2 m} + \frac{\omega^2 p_{0x}\sin\Omega\tau}{\Omega^3 m}$$

$$+ \frac{\omega p_{0y}}{\Omega^2 m}(1-\cos\Omega\tau) = (\text{now take } \Omega\to 0) =$$

$$= \Lambda\omega\frac{\mathcal{E}_0}{mc}\frac{\cancel{\Omega^2}\tau^3}{6} + \frac{p_{0x}}{\Omega^2 m}\underbrace{\left(\Lambda^2\tau - \omega^2\frac{\sin\Omega\tau}{\Omega}\right)}_{\underbrace{(\Lambda^2-\omega^2)}_{-\Omega^2}\tau + \frac{\omega^2}{6}\tau^3\Omega^2} =$$

$$+ \frac{\omega p_{0y}}{2m}\tau^2$$

$$= \frac{p_{0x}}{m}\tau - \frac{p_{0x}}{m}\frac{\omega^2\tau^3}{6} + \frac{\Lambda\omega\mathcal{E}_0}{mc}\frac{\tau^3}{6} + \frac{\omega p_{0y}\tau^2}{2m} =$$

remember that now $\Lambda=\omega$

$$= \frac{p_{0x}}{m}\tau + \frac{qEp_{0y}}{2m^2c}\tau^2 + \frac{\omega^2}{6m}\left(\frac{\mathcal{E}_0}{c} - p_{0x}\right)\tau^3$$

$$= \frac{p_{0x}}{m}\tau + \frac{qEp_{0y}}{2m^2c}\tau^2 + \frac{(qE)^2}{6(mc)^3}(\mathcal{E}_0 - cp_{0x})\tau^3. \quad (2')$$

And $$y(\tau) = \frac{\omega}{\Omega^2 mc}\left(\frac{\Lambda}{\omega}\mathcal{E}_0 - cp_{0x}\right)\left(\frac{1}{2}\Omega^2\tau^2\right) + \frac{p_{0y}}{m\Omega}(\Omega\tau)$$

$$= \frac{p_{0y}}{m}\tau + \frac{\omega}{2mc}(\mathcal{E}_0 - cp_{0x})\tau^2 =$$

$$= \frac{p_{0y}}{m}\tau + \frac{qE}{2(mc)^2}(\mathcal{E}_0 - cp_{0x})\tau^2. \quad (3')$$

Equs (1', 2', 3') are the solution for $E = CB$ case.

Suppose now that initially $\vec{p}_0 = 0 \Rightarrow \mathcal{E}_0 = mc^2 \Rightarrow$

$$ct = \frac{mc^2}{mc}\tau + \frac{(qE)^2}{6(mc)^3}mc^2\tau^3 = c\tau + \frac{1}{6}\left(\frac{qE}{mc}\right)^2 c\tau^3$$

$$t = \tau + \frac{1}{6}\left(\frac{qE}{mc}\right)^2\tau^3.$$

Next $x(\tau) = \dfrac{(qE)^2}{6(mc)^3} mc^2 \tau^3 = \dfrac{1}{6}\left(\dfrac{qE}{mc}\right)^2 c\tau^3.$

$$y(\tau) = \frac{qE}{2(mc)^2} mc^2 \tau^2 = \frac{qE}{2m}\tau^2.$$

Thus $\tau = \left(\dfrac{2my}{qE}\right)^{1/2} \Rightarrow x = \dfrac{1}{6}\left(\dfrac{qE}{mc}\right)^2 c\left(\dfrac{2my}{qE}\right)^{\frac{3}{2}}$

$$= \frac{1}{6}\left(\frac{qE}{mc}\right)^2 \left(\frac{mc}{qE}\right)^{\frac{3}{2}} (2y)^{3/2} \frac{c}{c^{3/2}} y^{3/2} =$$

$$= \frac{1}{6}\left(\frac{qE}{mc}\right)^{1/2}\left(\frac{2}{c}\right)^{1/2} 2y^{3/2} =$$

$$= \frac{1}{3}\left(\frac{2qE}{mc^2}\right)^{1/2} y^{3/2}.$$

$$x = \frac{1}{3}\left(\frac{2qE}{mc^2}\right)^{1/2} y^{3/2}.$$

OR $y = \left[3\left(\dfrac{mc^2}{2qE}\right)^{1/2}\right]^{2/3} x^{2/3} =$

$$= 3^{2/3}\left(\frac{mc^2}{2qE}\right)^{\frac{1}{3}} x^{\frac{2}{3}}.$$